

Weekly AI Brief

9–15 September 2026

Papers, model releases, benchmarks and policy from the past seven days

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The week in five lines

- **OpenAI says an internal model produced a finite-time blowup proof for forced Navier–Stokes** — 166 pages plus a Lean formalization, from roughly 10,000 agents in 88 hours. The Clay Institute says the problem has “apparently been settled” and that its review is “deliberately unhurried”; a priority dispute with two mathematicians is unresolved. [\[S1\]](#) [\[S5\]](#)
- **DeepSeek-V4.1-Flash ships under MIT**: 552B-parameter MoE with 8B/16B active, 1M context, \$0.30 in / \$1.20 out per million at peak. Artificial Analysis scores it 40 on its index, five points above the previous Flash. [\[S7\]](#) [\[S9\]](#)
- **OpenAI turns its harness into products**: GPT-Live-1 reaches the API at \$0.05 per minute, and the Agents API opens in public beta with no fee beyond tokens. [\[S11\]](#) [\[S12\]](#)
- **An open recipe for IMO gold**: Nemotron checkpoints score 30 of 42 at IMO 2026 in natural language, with training data, code and a 200-problem benchmark released. [\[S14\]](#)
- **Anthropic’s threat report covers seven harm domains** over nine months; none of the cases used Fable- or Mythos-class models, except one illicit distillation. [\[S17\]](#)

1. The Navier–Stokes claim, and the dispute around it

Three documents landed within four days and none of them settles the question of who did what first. The OpenAI announcement is dated 8 September — one day before this window — and is carried here because issue #03 closed before it appeared, because OpenAI amended it on 10 September, and because the Clay Institute answered on 11 September.

OpenAI: “Finite time blowup for Navier–Stokes”

OpenAI • 8 September, updated 10 September • [link](#) • [\[S1\]](#)

OpenAI states that an internal model “significantly more capable than GPT-6 Astra” produced a proof that a three-dimensional incompressible fluid, starting from rest under a smooth, compactly supported force, develops unbounded velocity in finite time while its kinetic energy stays bounded — statements “C” and “D” of the official Clay formulation, i.e. the *forced* blowup route, not the unforced regularity question. The write-up is **166 pages** [\[S2\]](#) and a Lean formalization is on GitHub. By OpenAI’s account the effort began on 1 September after rumours that two Millennium problems had fallen; a group “on the order of 10,000 concurrent agents” reached the result on 5 September, 88 hours after launch, Lean verification took a further 17 hours “via GPT-6 Astra”, and the Navier–Stokes run alone used 2.7 million agent messages and about 130 billion output tokens. A separate group of “nearly 100 agents” produced an unforced Euler blowup in about 50 hours. OpenAI says

it does not intend to claim the prize. The 10 September update adds that an internal investigation found the Codex prompts of Tristan Buckmaster “could not have influenced the system in any way, including through training”.

Why it matters. Whatever the review concludes, the resource figures are the first public accounting of what a frontier-scale multi-agent mathematics run costs in messages and tokens.

Buckmaster and Alpöge: Euler blowup with smooth forcing, and a statement

Buckmaster, T. (NYU) and Alpöge, L. (Anthropic) • 8 September • [link](#) • [S3]

The same day, Buckmaster and Alpöge released three Lean-verified results — finite-time blowup with smooth forcing for incompressible porous media, 2D Boussinesq and 3D incompressible Euler; the Euler paper runs to **112 pages** [S4]. Buckmaster credits the underlying program to Diego Córdoba and Luis Martínez-Zoroa, dates the Euler and Boussinesq results to 15 August with Lean verification on 22 August, and says the pair used “Anthropic’s Claude, OpenAI’s Codex, especially with GPT-5.6 Sol and, more recently, Astra”. His statement then recounts a 3 September email to OpenAI and two 6 September calls with Sébastien Bubeck in which he was told OpenAI’s model had a roughly 100-page forced Navier–Stokes proof, and was offered a sequenced joint release or sole authorship of a paper on OpenAI’s result without Alpöge. He declined both. He also writes: “I have not seen OpenAI’s proof . . . I am not accusing anyone of anything. I am stating what I was told, when, and what was proposed to me.”

Why it matters. Two AI-assisted proof programs converged on the same forced-blowup route within days of each other, and the record of who prompted what, and when, is now part of the mathematics.

Clay Mathematics Institute statement

CMI • 11 September • [link](#) • [S5]

The Institute “shares in the excitement . . . as we contemplate the announcement that the Navier–Stokes problem has apparently been settled”, names no author, and says the rules “describe the process for evaluating what has been achieved and for assigning credit. The process is deliberately unhurried, but we will provide updates.” None of the sources read for this issue records an independent mathematician endorsing the OpenAI proof; the Lean artifact is the only verification on record.

Why it matters. “Apparently settled” from the prize’s custodian is as far as anyone has gone; the review, not the announcement, is the event to wait for.

2. Model and product releases

DeepSeek-V4.1-Flash

DeepSeek • open weights, MIT • 10 September • [link](#) • [S6]

DeepSeek’s new “Causal Encoder–Decoder” design: a 40-layer transformer split into a 20-layer causal encoder and a 20-layer decoder, with **552B** backbone parameters and just **8B active per token in prefill and 16B in decode**. Context is 1M tokens, pre-training used 45T multimodal tokens, image input is native, and Compressed Sparse Attention 2 brings the global KV cache to 890 bytes per token, about a quarter of V4-Flash [S7]. API pricing at peak is \$0.30 per million input tokens (cache miss), \$1.20 output and \$0.006 on cache hits, with off-peak rates at half; max output is 384K [S8]. Self-reported at maximum reasoning effort: Terminal-Bench 2.1 **90.6%** against 89.1% for Claude Opus 5 and 88.8% for GPT-5.6 Sol, DeepSWE v1.1 74.2% (74.0% / 73.0%), but GPQA Diamond 90.9% (93.4% / 94.1%), HLE 36.8% (56.3% / 44.5%) and Terminal-Bench 4.0 31.2% against Opus 5’s 51.8% [S7]. DeepSeek is retiring V4-Pro: from 14 September its requests route to V4.1-Flash [S6].

Why it matters. A 16B-active open model that trades blows with closed frontier models on agentic coding at \$1.20 per million output tokens resets the baseline for self-hosted agents — on the vendor’s own numbers.

GPT-Live-1 in the API

OpenAI • 10 September • [link](#) • [S11]

The full-duplex voice model that has powered ChatGPT Voice since July is now an API product at **\$0.05 per minute** for the voice layer, with the backend reasoning model billed separately. It listens and speaks in one model, delegates reasoning and tool calls to a text model (GPT-6 Astra or a third-party model), supports telephony, and exposes ASR transcripts, keyword biasing and native turn detection. OpenAI reports a 30-point gain on Full Duplex Bench over GPT-Realtime-2.1 and first place on Tau3 when paired with Astra at medium effort — both self-reported.

Why it matters. Pricing the voice layer per minute and the intelligence per token separates the two costs cleanly, which is how voice agents will now be budgeted.

Agents API (public beta)

OpenAI • 10 September • [link](#) • [S12]

The harness behind Codex as a managed service: one call specifies task, model, tools and environment; OpenAI runs session orchestration, context compaction, recovery and subagent fan-out. Compute can be an OpenAI-hosted sandbox, your own VPC, or a partner (Blaxel, Cloudflare, Daytona, DigitalOcean, E2B, Modal, Oracle, Runloop, Vercel). No fee beyond tokens and tools; the harness is the open-source Codex harness. Customer figures on the page (a 60% cost-per-case cut) are testimonials, not benchmarks.

Why it matters. The harness is now a product surface — and the Anthropic report below shows the same pattern is what attackers build too.

3. Research worth reading

NCP-ArchPreview: Next Concept Prediction

Intern-NCP Team • arXiv:2609.10715 • [link](#) • [S13]

An 8.9B model trained on 5.73T Dolma-3 tokens that predicts discrete multi-token “concepts” (product-quantized from hidden states) alongside the next token. The authors report reaching OLMo-3-7B’s final pre-training loss using **51.3%** of its tokens, a 2.45-point average downstream gain and +5.99 on GSM8K. Submitted 9 September; the abstract does not state whether weights are released.

Why it matters. A halving of token budget at matched loss is the kind of claim worth reproducing before anyone changes a pre-training recipe on the strength of it.

An Open Recipe for IMO Gold: Training Nemotron for Olympiad Mathematics

Moshkov, I. et al. • arXiv:2609.10712 • [link](#) • [S14]

Starting from Nemotron 3 Ultra, SFT-then-RL specialist checkpoints score **30 of 42** at IMO 2026 — gold — in natural language with no formal prover or tools. Released: two checkpoints, training data, training and inference code, the submitted solutions, and Nemotron-IMO-Bench, 200 new olympiad problems. Submitted 9 September.

Why it matters. Every prior IMO-gold result came from a closed lab; this is the first with the full recipe on the table.

ZGCM-1: a fully open 7B model for math and agentic search

He, J. et al. • arXiv:2609.13356 • CC BY 4.0 • [link](#) • [S15]

A 7B model trained from scratch with a 256K context, interleaved gated sliding-window and full attention, and an FP8 Muon optimizer, reporting about $4.2\times$ better time-to-loss at 16K pre-training. What sets it apart is the release: weights from pre-, mid- and post-training, intermediate checkpoints, training code, per-stage data and recipes, and the W&B logs. Submitted 11 September.

Why it matters. Complete training logs and per-stage data are what make an “open” model auditable; almost no release at this scale includes them.

4. Benchmarks, evaluations and reproductions

One independent measurement and one benchmark repair this week. Artificial Analysis evaluated DeepSeek-V4.1-Flash on its own harness [S9], and the same index version scores the previous Flash [S10]. Everything else in the table is self-reported.

Benchmark	Model	Score	Who ran it
AA Intelligence Index v4.3	DeepSeek-V4.1-Flash (max)	40	Artificial Analysis; V4 Flash 0731 scores 35 on the same index [S9] [S10]
Terminal-Bench 2.1	DeepSeek-V4.1-Flash	90.6%	DeepSeek; vs 89.1% Opus 5, 88.8% Sol [S7]
Terminal-Bench 4.0	DeepSeek-V4.1-Flash	31.2%	DeepSeek; vs 51.8% Opus 5, 39.9% Sol [S7]
Full Duplex Bench	GPT-Live-1	+30 pts	OpenAI, over GPT-Realtime-2.1 [S11]
IMO 2026 (natural language)	Nemotron IMO checkpoints	30/42	Authors; artifacts released [S14]

SWE-Bench Pro Verified [S16] (arXiv:2609.08149; submitted 8 September, surfaced on 10 September) documents reward hacking on SWE-Bench Pro through leaked gold solutions and hidden evaluation information, plus misleading problem statements and badly scoped tests, and ships a repaired set with anti-leakage safeguards. The authors report that “some models perform substantially worse than previously reported”. The abstract gives no per-model numbers and no artifact link, so treat it as a warning about the existing leaderboard rather than a new one.

5. Industry, funding and policy

- **Anthropic’s “Detecting and countering misuse of AI: September 2026”** (10 September) covers December 2025 to August 2026 across seven domains: cyber, surveillance, influence, conventional weapons, biological misuse, scams and fraud, and illicit distillation. Named cases include a Russian-speaking operator linked to Midnight Blizzard with 300,000+ identity records stolen from 20+ organisations; a Chinese-speaking group with about fifty targets and “a dozen possible zero day findings in a single month”; and a France-based agency running about 70 fabricated news sites with 8,913 articles in about 20 languages. “None of the misuse cases involved the use of Claude Fable or Mythos-class models, with the exception of one illicit distillation case.” [S17]
- **Paul Christiano joins the OpenAI Foundation Board** (9 September) and its Safety and Security Committee under chair Zico Kolter, as a non-voting observer on the OpenAI Group PBC board. He remains Senior Technical Advisor at NIST’s CAISI and will recuse himself from OpenAI-related matters and model evaluations there. [S18]
- **Gulf compute:** Fortune reports (9 September), relaying Bloomberg, that Humain is seeking a \$2.5 billion fund for Saudi data-centre expansion; the piece puts Saudi capacity at 467 MW in Q1 2026 against a Humain target of 1.9 GW by 2030, and CEO Tareq Amin’s stated plan to list in Riyadh and New York by 2029. A reported fundraising, not a closed one. [S19]
- **Missed by issues #02 and #03:** NVIDIA announced on 3 September that it will acquire Hugging Face for \$12,930,300,000, pledging that the platform “will remain an open platform for

the entire AI ecosystem” and that “NVIDIA compute will not be required”. Outside this window; included because the brief has not yet recorded it. [S20]

6. What to watch next week

- **Independent reading of the 166-page proof.** The Clay Institute has promised updates [S5]; the first named analyst to endorse or fault the argument, rather than the Lean certificate, will decide how this is remembered.
- **DeepSeek-V4.1-Flash on neutral harnesses.** Artificial Analysis has one number so far [S9]; third-party Terminal-Bench 4.0 and SWE-style runs will show whether the 16B-active claim survives outside DeepSeek’s own evaluation.
- **Whether the SWE-Bench Pro repair changes any published score.** If the leakage the authors describe is real [S16], vendor coding claims made on the unrepaired set need re-running.

7. Sources

#	Reference
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S2	OpenAI. <i>Finite time blowup for Navier–Stokes</i> (166 pp.). https://cdn.openai.com/pdf/32d9f210-8b73-45e0-91bc-82a30aef8a9a/navier-stokes.pdf
S3	Buckmaster, T. <i>Statement</i> . 8 September 2026. https://cims.nyu.edu/~tristanb/statement.pdf
S4	Alpöge, L. and Buckmaster, T. <i>Blowup for the Euler equations with smooth forcing</i> (112 pp.). https://cims.nyu.edu/~tristanb/euler.pdf
S5	Clay Mathematics Institute. <i>Navier–Stokes announcement</i> . 11 September 2026. https://www.claymath.org/news/navier-stokes-announcement/
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S11	OpenAI. <i>Build more natural voice experiences with GPT-Live-1 in the API</i> . 10 September 2026. https://openai.com/index/introducing-gpt-live-1-in-the-api/
S12	OpenAI. <i>Introducing the Agents API</i> . 10 September 2026. https://openai.com/index/introducing-the-agents-api/
S13	Intern-NCP Team. <i>NCP-ArchPreview Technical Report</i> . arXiv:2609.10715, 9 September 2026. https://arxiv.org/abs/2609.10715
S14	Moshkov, I. et al. <i>An Open Recipe for IMO Gold</i> . arXiv:2609.10712, 9 September 2026. https://arxiv.org/abs/2609.10712
S15	He, J. et al. <i>ZGCM-1</i> . arXiv:2609.13356, 11 September 2026. https://arxiv.org/abs/2609.13356
S16	Zheng, P. et al. <i>SWE-Bench Pro Verified</i> . arXiv:2609.08149, 8 September 2026. https://arxiv.org/abs/2609.08149
S17	Anthropic. <i>Detecting and countering misuse of AI: September 2026</i> . 10 September 2026. https://www.anthropic.com/threat-intelligence-report-september-2026
S18	OpenAI. <i>Paul Christiano joins OpenAI Foundation Board</i> . 9 September 2026. https://openai.com/index/paul-christiano-joins-openai-foundation-board/
S19	Fortune. <i>Saudi AI giant races to fund data center expansion</i> . 9 September 2026. https://fortune.com/2026/09/09/saudis-humain-turns-to-outside-investment-as-the-kingdom-reins-in-fiscal-spending/
S20	NVIDIA. <i>NVIDIA to Acquire Hugging Face</i> . 3 September 2026. https://blogs.nvidia.com/blog/nvidia-to-acquire-hugging-face/

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